Locally Convex Topological Vector Spaces

Definitions and Examples

Definition: A topological vector space (TVS) is a vector space X with a topology $\mathcal{T}$ such that, with respect to this topology, the maps

(i) $X \times X \to X$, $(x, y) \mapsto x + y$

(ii) $\mathbb{F} \times X \to X$, $(\alpha, x) \mapsto \alpha x$

are continuous with respect to $\mathcal{T}$.

Example: If $\mathcal{P}$ is a family of seminorms on X, and $\mathcal{T}$ is the topology on X with subbasic sets $U(x_0, p, \varepsilon) = \{x \in X : p(x - x_0) < \varepsilon\}$, then $(X, \mathcal{T})$ is a topological vector space.

Definition: A locally convex topological vector space (LCS) is a TVS equipped with a family of seminorms $\mathcal{P}$ that satisfy

$$\bigcap_{p \in \mathcal{P}} \{x : p(x) = 0\} = \{0\}.$$

That is, the family $\mathcal{P}$ separates the points of X.

Proposition: Let X be a TVS, and let p be a seminorm on X. The following are equivalent:

(a) p is continuous;

(b) $\{x \in X : p(x) < 1\}$ is open;

(c) $0 \in \{x \in X : p(x) < 1\}^{\circ}$;

(d) $0 \in \{x \in X : p(x) \leq 1\}^{\circ}$;

(e) p is continuous at 0;

(f) there is a continuous seminorm q on X such that $p \leq q$.

Proof. We see immediately that (a) implies (b) since the set is equal to the inverse image of the open subset $[0, 1) \subseteq [0, \infty)$. Similarly, if $p^{-1}([0, 1))$ is open, then it is equal to its own interior, and since $0 \in p^{-1}([0, 1))$, it follows that there is a subbasic open neighborhood of 0 contained in $p^{-1}([0, 1))$, giving (c). Similarly, we obtain (d) by the fact that $0 \in p^{-1}([0, 1))^{\circ} \subseteq p^{-1}([0, 1])^{\circ}$.

Next, we see that (d) implies that for every $\varepsilon > 0$, $0 \in p^{-1}([0, \varepsilon])^{\circ}$, so if $(x_i)_i$ is a net in X converging to 0 and $\varepsilon > 0$, there is some i_0 such that for all $i \geq i_0$, $x_i \in \{x : p(x) \leq \varepsilon\}$. That is, $p(x_i) \leq \varepsilon$ for all $i \geq i_0$, whence p is continuous at 0. This gives (e).

If p is continuous at 0, then if $x_i \to x$ is a convergent net in X, the reverse triangle inequality gives $|p(x) - p(x_i)| \leq p(x_i - x)$, whence $p(x_i) \to p(x)$, so p is continuous. This gives that (e) implies (a).

Taking $q = p$ yields (a) implies (f). We will show that (f) implies (e).If $x_i \to 0$ in X, then $q(x_i) \to 0$, but since $0 \leq p(x_i) \leq q(x_i)$, it follows that $p(x_i) \to 0$. □

Example:

(a) Let X be a completely regular topological space, and let $C(X)$ be the set of all continuous functions from X to $\mathbb{F}$. If K is compact in X, we may define a seminorm $p_K(f) = \sup_{x \in K} |f(x)|$. Then,

$$\{p_K : K \subseteq X \text{ compact}\}$$

forms a separating family of seminorms for $C(X)$.

(b) Let G be an open subset of $\mathbb{C}$. If $H(G)$ is the set of holomorphic functions on G, then the same seminorms $\{p_K : K \subseteq G \text{ compact}\}$ forms a locally compact space whose topology is uniform convergence on compact subsets.

(c) Let X be a normed vector space. For each $\varphi \in X^*$, let $p_{\varphi}(x) = \left|\varphi(x)\right|$. Then, $\mathcal{P} = \left\{p_{\varphi} : \varphi \in X^*\right\}$ forms a separating family of seminorms for X. The topology these seminorms induce is called the *weak topology* on X. It is often denoted $\sigma(X, X^*)$.

(d) If X is a normed space, then for each $x \in X$, we may define a seminorm $p_x \colon X^* \to [0,\infty)$ by taking $p_x(\varphi) = \left|\varphi(x)\right|$. This defines a separating family of seminorms $\mathcal{P} = \{p_x : x \in X\}$, and the topology on X^* induced by these seminorms is called the weak*, or w^*, topology. It is often denoted $\sigma(X^*, X)$.

Next, we focus on an examination of the "convexity" part of the notion of locally convex topological vector spaces.

Definition: A subset $A \subseteq X$ is said to be convex if for all $v, w \in A$, the segment

$$[v,w] := \{tv + (1-t)q : 0 \leq t \leq 1\}$$

is contained in A.

Note that the intersection of any family of convex subsets is convex.

Definition: If $A \subseteq X$ is any subset, then the *convex hull* of A, denoted $\operatorname{conv}(A)$, is the intersection of all the convex subsets that contain A. If X is a TVS, the *closed convex hull* of A, denoted $\overline{\operatorname{conv}}(A)$, is the intersection of all the closed convex sets that contain A.

Given any linear map $T \colon X \to Y$ of real vector spaces, and $C \subseteq X$ a convex subset of Y, then $T^{-1}(C)$ is convex in X. If X^* is the dual of any complex vector space, then since X is also a real vector space, it then follows that the sets

$$\begin{aligned} A_0 &= \{x : |f(x)| \leq 1\} \\ A_1 &= \{x : \operatorname{Re}(f(x)) \leq 1\} \\ A_3 &= \{x : \operatorname{Re}(f(x)) > 1\} \end{aligned}$$

are all convex.

Proposition: Let X be a TVS, and let A be a convex subset of X. Then,

(a) $\overline{A}$ is convex;

(b) if $a \in A^{\circ}$ and $b \in \overline{A}$, then

$$[a,b) := \{tb + (1-t)a : 0 \leq t < 1\}$$

is contained in A°.

Proof. Let $a \in A$, $b \in \overline{A}$, and let $0 \leq t \leq 1$. Let $(x_i)_i$ be a net in A such that $x_i \to b$. Then, $tx_i + (1-t)a \to tb + (1-t)a$. In particular, this means that if $b \in \overline{A}$ and $a \in A$, then $[a,b] \subseteq \overline{A}$. So, given any $a, b \in \overline{A}$, we have that the net $(x_i)_i \to b$ is such that $[x_i, a] \subseteq \overline{A}$, so it follows that $[b,a] \subseteq \overline{A}$, whence $\overline{A}$ is convex.

For (b), fix $0 < t < 1$, and set $c = tb + (1-t)a$ for some $a \in A^{\circ}$ and $b \in \overline{A}$. Since $a \in A^{\circ}$, there is an open neighborhood V of 0 such that $a + V \subseteq A$. Therefore, for any $d \in A$, we have

$$\begin{aligned} A &\supseteq td + (1-t)(a+V) \\ &= t(d-b) + tb + (1-t)(a+V) \\ &= (t(d-b) + (1-t)V) + c. \end{aligned}$$

We hope to find some $d \in A$ such that $0 \in t(d-b) + (1-t)V$, which we call U. The preceding inclusion then shows that $c \in A^{\circ}$, since U is an open neighborhood of 0 with $c + U \subseteq A$.

Finding such a d is equivalent to finding d such that $0 \in \frac{1}{t}(1-t)V + (d-b)$, or $d \in b - \frac{1}{t}(1-t)V$,. Yet, we know that $0 \in -\frac{1}{t}(1-t)V$ by the assumption that V is an open neighborhood of 0. Since $b \in \overline{A}$, it follows that $b - \frac{1}{t}(1-t)V \cap A \neq \emptyset$ by the definition of closure, whence such a $d \in A$ exists. $\square$

Corollary: If $A \subseteq X$, then $\overline{\operatorname{conv}}(A) = \overline{\operatorname{conv}(A)}$.

Definition: A set $A \subseteq X$ is called balanced if for any $|\alpha| \leq 1$ and any $x \in A$, we have $\alpha x \in A$.

We say a set A is *absorbing* if for any $x \in X$, there is some $\varepsilon > 0$ such that $tx \in A$ for all $0 \leq t < \varepsilon$. We say A is absorbing at a if for any $x \in X$, there is $\varepsilon > 0$ such that $a + tx \in A$ for all $0 \leq t < \varepsilon$.

Given a seminorm p on V, the set $V = \{x : p(x) < 1\}$ is a convex balanced set that is absorbing at each of its points. We will show that the converse actually holds — that is, a vector space is a LCS if and only if the set of all open, convex, balanced subsets of X forms a neighborhood basis at 0.

Proposition: Suppose X is a vector space over F, and V is a nonempty convex balanced set that is absorbing at each of its points. Then, there is a unique seminorm p on X such that

$$V = \{x \in X : p(x) < 1\}.$$

Proof. Let

$$p(x) = \inf\{t \geq 0 : x \in tV\}.$$

Since V is absorbing, it follows that $X = \bigcup_{n=1}^{\infty} nV$, so this is a well-defined quantity. Furthermore, $p(0) = 0$.

To show that $p(\alpha x) = |\alpha| p(x)$, we suppose $\alpha \neq 0$, and since V is balanced,

$$\begin{aligned} p(\alpha x) &= \inf\{t \geq 0 : \alpha x \in tV\} \\ &= \inf\left\{t \geq 0 : x \in \frac{t}{\alpha} V\right\} \\ &= \inf\left\{t \geq 0 : x \in \frac{t}{|\alpha|} V\right\} \\ &= |\alpha| \inf\left\{\frac{t}{|\alpha|} : x \in \frac{t}{|\alpha|} V\right\} \\ &= |\alpha| p(x). \end{aligned}$$

We now observe that if $\alpha, \beta \geq 0$, then for any $a, b \in V$, we have

$$\begin{aligned} \alpha a + \beta b &= (\alpha + \beta)\left(\frac{\alpha}{\alpha + \beta} a + \frac{\beta}{\alpha + \beta} b\right) \\ &\in (\alpha + \beta) V, \end{aligned}$$

since V is convex. For any $x, y \in X$, take $p(x) = \alpha$ and $p(y) = \beta$, and let $\delta > 0$. Then, $x \in (\alpha + \delta)V$ and $y \in (\beta + \delta)V$ by the definition of the infimum, so that $x + y \in (\alpha + \beta + 2\delta)V$. Therefore, we have $p(x + y) \leq \alpha + \beta + 2\delta$, which holds for all $\delta > 0$, so that $p(x + y) \leq p(x) + p(y)$.

Finally, we will have to show that $V = \{x : p(x) < 1\}$. If $p(x) = \alpha < 1$, then for any $\alpha < \beta < 1$, we have $x \in \beta V \subseteq V$, which emerges from the fact that V is balanced. Thus, $V \supseteq \{x : p(x) < 1\}$. If $x \in V$, then $p(x) \leq 1$, and since V is absorbing at x, there is some $\varepsilon > 0$ such that for $0 < t < \varepsilon$, we have $y = x + tx \in V$. Yet, this means $x = \frac{1}{1+t} y$ with $y \in V$, so that $p(x) = \frac{1}{1+t} p(y) \leq \frac{1}{1+t} < 1$. □

The seminorm p is called the Minkowski function for V, or the *gauge* of V.

Proposition: Let X be a TVS, and let $\mathcal{U}$ be the collection of open, convex, balanced subsets of X. Then, X is locally convex if and only if $\mathcal{U}$ is a basis for the neighborhood system at 0.

Proof. Suppose X is locally convex with topology defined by the family of seminorms $\mathcal{P}$. Then, we observe that each of the subbasic open neighborhoods of 0,

$$\{U(0, p, \varepsilon) : \varepsilon > 0, p \in \mathcal{P}\},$$

are open, convex, and balanced; since the set of all finite intersections of all these subbasic open neighborhoods yields a neighborhood basis for 0, and finite intersections of open, balanced, convex subsets are open, balanced, and convex, so it follows that $\mathcal{U}$ contains the set of all finite intersections of all these subbasic open neighborhoods, whence $\mathcal{U}$ contains a basis for the neighborhood system at 0. In particular, this means $\mathcal{U}$ also must generate the neighborhood system at 0, so $\mathcal{U}$ forms a basis for the neighborhood system at 0.

Now, suppose $\mathcal{U}$ forms a basis for the neighborhood system at 0. Observe that an open, convex, balanced subset of V contains 0, and is itself absorbing. We claim that the family of Minkowski functionals

$$\mathcal{P} = \{p_V(\cdot - z) : z \in X, V \in \mathcal{U}\}$$

generates the topology for X. □

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References

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